

Multiple Choice

1. **Answer:** D

Options: A) 53; B) 29; C) 50; D) 100

Note: Correct option: D - 100

2. **Answer:** A

Options: A) Temporary inconsistencies among views of a file by different machines can result; B) The file system is likely to be corrupted when a computer crashes; C) A much higher amount of network traffic results; D) Caching makes file migration impossible

Note: Correct option: A - Temporary inconsistencies among views of a file by different machines can result

3. **Answer:** A

Options: A) $O(1)$; B) $O(n)$; C) $O(n^2)$; D) $O(\log n)$

Note: Correct option: A - $O(1)$

4. **Answer:** A

Options: A) Recursive procedures; B) Arbitrary goto's; C) Two-dimensional arrays; D) Integer-valued functions

Note: Correct option: A - Recursive procedures

5. **Answer:** A

Options: A) one in which each main memory word can be stored at any of 3 cache locations; B) effective only if 3 or fewer processes are running alternately on the processor; C) possible only with write-back; D) faster to access than a direct-mapped cache

Note: Correct option: A - one in which each main memory word can be stored at any of 3 cache locations

True / False

6. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: ' $O((\log N)^2)$ '.

7. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

8. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

9. **Answer:** True
Options: A) True; B) False
Note: LLM-generated true statement based on MCQ.
10. **Answer:** True
Options: A) True; B) False
Note: LLM-generated true statement based on MCQ.

Long Form Response

11. **Answer:** `def is_Two_Alter(s): | return False | return False | return True`
Note: Students should provide a detailed explanation referencing algorithm steps.
12. **Answer:** `def move_num(test_str): | return (res)`
Note: Students should provide a detailed explanation referencing algorithm steps.

End of Answer Key